Fast cube roots in $\mathbb{F}_{p^2}$ via the algebraic torus

“Project, Rotate, Recover”

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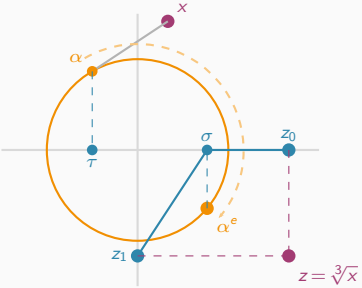
Consensys – Linea



Why $\mathbb{F}_{p^2}$ cube roots?

Applications of $\mathbb{F}_{p^2}$ cube roots

- **Hash-to-curve** (Icart and Koshelev maps [Ica09, Kos21b])

  Koshelev: "does not know how to express $\sqrt[3]{\cdot}$ in $\mathbb{F}_{p^2}$ through $\mathbb{F}_p$-roots"

- **Point decompression** on $j=0$ curves ($y^2 = x^3 + b$ over $\mathbb{F}_{p^2}$) / $\mathbb{G}_2$ subgroups
  - **Highly 2-adic fields (SNARK)**: cube root cheaper than square root
  - **Batch decompression (via quotient varieties [Kos21a])**: requires cube roots

- **3-isogeny walks [COR25]**: bottleneck = one $\mathbb{F}_{p^2}$ cube root per step

3-isogeny walk

$$E_0 \xrightarrow{\sqrt[3]{\cdot}} E_1 \xrightarrow{\sqrt[3]{\cdot}} E_2 \xrightarrow{\sqrt[3]{\cdot}} \cdots \xrightarrow{\sqrt[3]{\cdot}} E_m$$

bottleneck: one $\mathbb{F}_{p^2}$ cube root per step

Application [NO24]	Security	Walk length	Field
QFESTA-128	128-bit	$m = 162$	$\mathbb{F}_{p^2}$
QFESTA-192	192-bit	$m = 243$	$\mathbb{F}_{p^2}$
QFESTA-256	256-bit	$m = 324$	$\mathbb{F}_{p^2}$

This talk

1. Reduce $\mathbb{F}_{p^2}$ cube root to $\mathbb{F}_p$ operations
   via the **algebraic torus** $\mathbb{T}_2(\mathbb{F}_p)$

2. **1.6–2.3**$\times$ faster than best prior (direct) method

Roots in $\mathbb{F}_p$

Square root in $\mathbb{F}_p$

- Simple cases via direct exponentiation:

  $p \equiv 3 \pmod 4$: $\sqrt{a} = a^{(p+1)/4}$, $\quad p \equiv 5 \pmod 8$: Atkin's formula, $\quad \ldots$

- General($v_2(p-1) \geq 3$): **Tonelli–Shanks**

  Exponentiation + discrete-log correction in 2-Sylow subgroup

- Cost $\approx 1.5 \log_2(p)\, \mathbf{M}_p$ (via square-and-multiply)
- Cost $\approx 1.2 \log_2(p)\, \mathbf{M}_p$ (via addition chain)

Cube root in $\mathbb{F}_p$

- Simple cases via direct exponentiation:

  $p \equiv 2 \pmod 3$: $\sqrt[3]{a} = a^{(2p-1)/3}$

  $p \equiv 4 \pmod 9$: $\sqrt[3]{a} = a^{(8p-5)/9}$

  $p \equiv 7 \pmod 9$: $\sqrt[3]{a} = a^{(p+2)/9}$

  $p \equiv 10 \pmod{27}$: $\sqrt[3]{a} = \zeta \cdot a^{(2p+7)/27}$

  $p \equiv 19 \pmod{27}$: $\sqrt[3]{a} = \zeta \cdot a^{(p+8)/27}$

- General ($v_3(p-1) \geq 3$): **Adleman–Manders–Miller [AMM77]**

  Exponentiation + discrete-log correction in 3-Sylow subgroup

- Cost $\approx 1.5 \log_2(p)\ \mathbf{M}_p$ (via square-and-multiply)
- Cost $\approx 1.2 \log_2(p)\ \mathbf{M}_p$ (via addition chain)

Square root vs. cube root in $\mathbb{F}_p$

v		**Square root** ($v_2(p-1)$)	**Cube root** ($v_3(p-1)$)
1	Formula	1 exp $(p \equiv 3 \bmod 4)$	1 exp $(p \equiv 4, 7 \bmod 9)$
2	Formula	1 exp $(p \equiv 5 \bmod 8)$	1 exp $(p \equiv 10, 19 \bmod 27)$
≥ 3	Algorithm	Tonelli–Shanks (DL in 2-Sylow)	AMM (DL in 3-Sylow)

All practical cryptographic primes have $v_3(p-1) \leq 2$

$\Rightarrow$ cube root in $\mathbb{F}_p$ = single exponentiation

From $\mathbb{F}_p$ to $\mathbb{F}_{p^2}$

$\mathbb{F}_{p^2}$ arithmetic recap

$$\mathbb{F}_{p^2} = \mathbb{F}_p[u]/(u^2 - \beta), \qquad \mathrm{N}(x) = x \cdot \bar{x} \in \mathbb{F}_p$$

- Conjugation: $\overline{x_0 + x_1 u} = x_0 - x_1 u$
- Frobenius: $x^p = \bar{x}$
- $\mathbf{M}_{p^2} = 3\,\mathbf{M}_p$ (Karatsuba)
- $\mathbf{S}_{p^2} = 2\,\mathbf{M}_p$ (Complex)

$\sqrt{\cdot}$ in $\mathbb{F}_{p^2}$ reduces to $\mathbb{F}_p$

Key identity: $b = x^{(p+1)/2}$ satisfies $b^2 = \mathrm{N}(x) \in \mathbb{F}_p$, so $b \in \mathbb{F}_p$

$$x \in \mathbb{F}_{p^2} \xrightarrow{x \cdot \bar{x}} \mathrm{N}(x) \in \mathbb{F}_p \xrightarrow{\mathbb{F}_p \text{ sqrt}} \sqrt{\mathrm{N}(x)} \in \mathbb{F}_p \xrightarrow{\text{lift}} z \in \mathbb{F}_{p^2}$$

$\Rightarrow$ $\mathbb{F}_{p^2}$ square root reduces to **one** $\mathbb{F}_p$ **square root** + cheap recovery

$\sqrt[3]{\cdot}$ in $\mathbb{F}_{p^2}$: why the same approach fails

Try $\gamma = x^{(p+1)/3}$: $\quad \gamma^3 = \mathrm{N}(x) \in \mathbb{F}_p \quad$ but $\gamma \notin \mathbb{F}_p$ in general!

$$x \in \mathbb{F}_{p^2} \xrightarrow{x \cdot \bar{x}} \mathrm{N}(x) \in \mathbb{F}_p \xrightarrow{\mathbb{F}_p \text{ cbrt}} \sqrt[3]{\mathrm{N}(x)} \in \mathbb{F}_p \overset{\times \text{ lift?}}{\dashrightarrow} z \in \mathbb{F}_{p^2}$$

Casus irreducibilis: the cubic for z_0 over $\mathbb{F}_p$ has no $\mathbb{F}_p$-radical solution

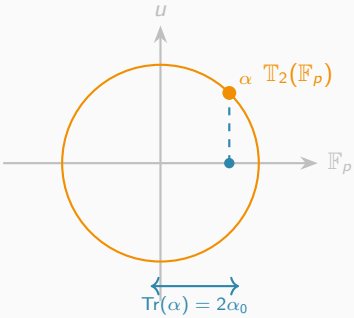
$\Rightarrow$ *Need a different approach*

Our algorithm

The algebraic torus $\mathbb{T}_2(\mathbb{F}_p)$

$$\mathbb{T}_2(\mathbb{F}_p) = \{\alpha \in \mathbb{F}_{p^2}^* : \mathsf{N}(\alpha) = 1\}, \qquad |\mathbb{T}_2| = p + 1$$

- $\alpha^p = \bar{\alpha} = \alpha^{-1}$
- α determined by $\mathsf{Tr}(\alpha) = 2\alpha_0 \in \mathbb{F}_p$



Key observation

1. $x^{p-1} \in \mathbb{T}_2(\mathbb{F}_p)$ (since $\mathrm{N}(x^{p-1}) = 1$)
2. $\gcd(3, p+1) = 1$ for $p \equiv 1 \pmod 3$

   $\Rightarrow$ **cube map is a bijection** on $\mathbb{T}_2(\mathbb{F}_p)$
3. Cube root on torus $=$ exponentiation by $e = 3^{-1} \bmod (p+1)$
4. **Connection to Lucas V-sequence:**

   $\alpha \in \mathbb{T}_2$ is determined by $\tau = \mathrm{Tr}(\alpha) \in \mathbb{F}_p$, and

   $$\mathrm{Tr}(\alpha^e) = V_e(\tau) \quad \text{(Lucas identity)}$$

   $\Rightarrow$ compute α^e via $V_e(\tau)$ **entirely in** $\mathbb{F}_p$

$$\sigma = V_e(\tau), \quad \tau = \mathrm{Tr}(\alpha), \quad e = 3^{-1} \bmod (p+1)$$

Algorithm overview

projection to $\mathbb{T}_2(\mathbb{F}_p)$

exponentiation in $\mathbb{T}_2(\mathbb{F}_p)$

recovery

$$x = x_0 + x_1 u \in \mathbb{F}_{p^2} \longrightarrow \begin{aligned} \tau &= \mathrm{Tr}(x^{p-1}) \\ &= 2(x_0^2 - |\beta| x_1^2) \cdot N(x)^{-1} \end{aligned} \longrightarrow \begin{aligned} \sigma &= V_e(\tau) \\ e &= 3^{-1} \bmod (p+1) \end{aligned} \longrightarrow \begin{aligned} z_0 &= \frac{x_0}{\sqrt[3]{\mathrm{N}(x)}\,(\sigma - 1)} \\ z_1 &= \frac{x_1}{\sqrt[3]{\mathrm{N}(x)}\,(\sigma + 1)} \end{aligned} \longrightarrow z = z_0 + z_1 u \in \mathbb{F}_{p^2}$$

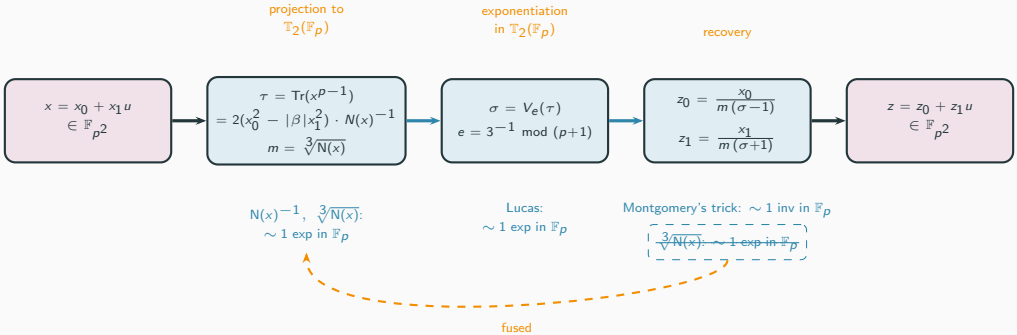
$\mathrm{N}(x)^{-1}$: ~ 1 inv in $\mathbb{F}_p$

Lucas: ~ 1 exp in $\mathbb{F}_p$

$\sqrt[3]{\mathrm{N}(x)}$: ~ 1 exp in $\mathbb{F}_p$

$x_0 / \cdot$, $x_1 / \cdot$: ~ 2 inv in $\mathbb{F}_p$

Algorithm overview (optimized)



Fused $\sqrt[3]{N(x)}$ and $N(x)^{-1}$

Let $n = \mathrm{N}(x)$. Write $p - r = d \cdot h$ with $d \in \{9, 27\}$ and $r \in \{4, 7, 10, 19\}$.

A single addition chain computes $t = n^h$, then:

$$m = n \cdot t^s \quad \text{(cube root)}, \qquad n^{-1} = m^a \cdot t^b \quad \text{(Fermat)}$$

where $s \in \{1, 2\}$, $a = r - 2$, $b = d - as$.

Prime	p mod 9	$(r,\ s)$	Saves
p-p377	7	(7, 4)	$\sim 50\,\mathbf{M}_p$ ($\approx$ 1 inversion)
i-p381	4	(4, 5)	

$\Rightarrow$ Cube root **and** inverse of $\mathrm{N}(x)$ from one addition chain

Lucas V-sequence on the torus

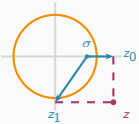
$$V_0 = 2, \quad V_1 = \tau$$

$$V_{2k} = V_k^2 - 2, \qquad V_{2k+1} = V_k \cdot V_{k+1} - \tau$$

- Cost: $1.8\,\mathbf{M}_p$ per bit $\quad (1\,\mathbf{M}_p + 1\,\mathbf{S}_p$, with $\mathbf{S}_p = 0.8\,\mathbf{M}_p)$
- Binary ladder: tight, branch-free, constant-time
- Total: $\approx 1.8\,\ell\,\mathbf{M}_p$ where $\ell = \log_2(p)$

Lucas dominates cost: 55–60% of total

Recovery & verification



Recovery:

$$z_0 = \frac{x_0}{m(\sigma - 1)}, \qquad z_1 = \frac{x_1}{m(\sigma + 1)}$$

Single $\mathbb{F}_p$ inversion via Montgomery's trick: $5\,\mathbf{M}_p + 1\,\mathbf{I}_p$

Verify and adjust (when $p \not\equiv 7 \pmod 9$)**:**

Let $\zeta_3 \in \mathbb{F}_p$ be a primitive cube root of unity ($\zeta_3^3 = 1$, $\zeta_3 \neq 1$).

- Check $z^3 = x$ ($1\,\mathbf{S}_{p^2} + 1\,\mathbf{M}_{p^2} = 5\,\mathbf{M}_p$)
- If not, multiply by precomputed ζ_3 or ζ_3^2 ($2\,\mathbf{M}_p$)

Cost comparison ($\ell = \log_2 p$)

Method	Cost	Ratio
Direct (addition chain in $\mathbb{F}_{p^2}$)	$\sim 5.3\ell\ \mathbf{M}_p$	$1.9\times$
Frobenius 1-bit	$\sim 4.25\ell\ \mathbf{M}_p$	$1.5\times$
Frobenius 2-bit	$\sim 3.4\ell\ \mathbf{M}_p$	$1.2\times$
Torus	$\sim 2.8\ell + 70\ \mathbf{M}_p$	$\mathbf{1}\times$

Structural advantage: all operations in $\mathbb{F}_p$ ($3\times$ cheaper than $\mathbb{F}_{p^2}$)

Benchmarks

Target primes

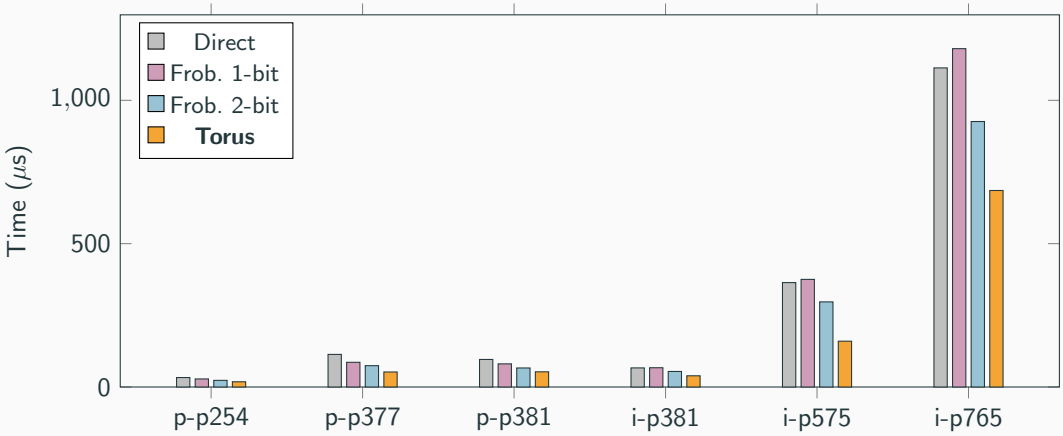
Prime	$\log_2 p$	$p \bmod 9$	β	**Source**
p-p254	254	1	-1	Pairing [BCTV14] (Ethereum)
p-p377	377	7	-5	Pairing [BCG+20] (Linea)
p-p381	381	1	-1	Pairing [Bow17] (Ethereum)
i-p381	381	4	-1	Isogeny [COR25]
i-p575	575	4	-1	Isogeny [COR25]
i-p765	765	4	-1	Isogeny [COR25]

Benchmark results

AMD EPYC 9R14, Go 1.24, median of 4 runs

Prime	Direct (μs)	Frob. 1 (μs)	Frob. 2 (μs)	Torus (μs)	vs. Direct	vs. Frob. 2
p-p254	33.0	28.1	23.2	**18.2**	1.81×	1.27×
p-p377	113.7	86.2	74.4	**52.4**	2.17×	1.42×
p-p381	96.1	80.7	66.4	**53.0**	1.81×	1.25×
i-p381	66.6	67.2	54.2	**39.1**	1.70×	1.39×
i-p575	364.2	375.4	296.9	**159.8**	2.28×	1.86×
i-p765	1113.1	1180.2	926.1	**685.5**	1.62×	1.35×

Visual comparison



Constant-time & conclusion

Constant-time considerations

All methods: fixed operation sequence, CT w.r.t. input

VerifyAndAdjust: input-dependent branch (multiply by ζ_3^k)

Application	**Secret input?**	**CT needed?**
Hash-to-curve	Yes	Yes
Point decompression	No	No
3-isogeny walk (CSIDH)	Yes	Yes

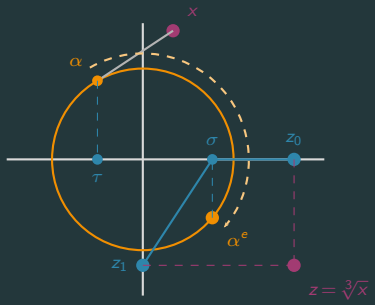
CT VerifyAndAdjust via conditional select ($\sim 6\,\mathbf{M}_p$ overhead)

Conclusion

- $\mathbb{F}_{p^2}$ cube root $\longrightarrow \mathbb{F}_p$ via **algebraic torus** + **Lucas sequences**
- **1.6–2.3**$\times$ vs. direct, **1.2–1.9**$\times$ vs. Frobenius 2-bit
- Extension to $p \equiv 2 \pmod 3$ discussed in paper
- Extension to $\mathbb{F}_{p^{2k}}$ + GLV-like optimization discussed in paper
- Extension to odd k-th roots discussed in paper
- Code: `github.com/yelhousni/fp2-cbrt`

Thank you!

github.com/yelhousni/fp2-cbrt



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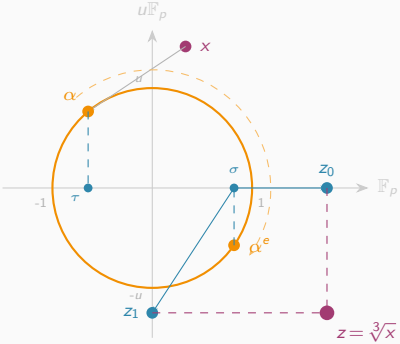
Appendix: Visual – $p \equiv 1 \pmod 3$



$\mathbb{T}_2(\mathbb{F}_p)$, $|\mathbb{T}_2| = p+1$

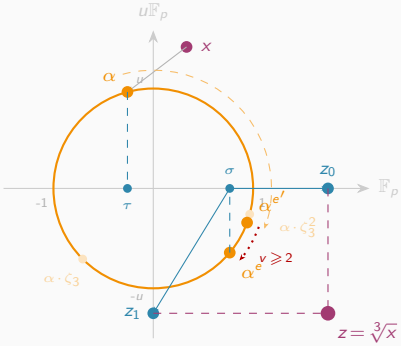
$\gcd(3,\ p+1) = 1$

$\Rightarrow$ cube map **bijective**

- Project: $\alpha = x^{p-1}$
- Rotate: $e = 3^{-1} \bmod (p+1)$
- Recover: $z_0,\ z_1 \in \mathbb{F}_p$

ζ_3-adjust if $p \not\equiv 7 \pmod 9$

Appendix: Visual – $p \equiv 2 \pmod 3$



$\mathbb{T}_2(\mathbb{F}_p)$, $|\mathbb{T}_2| = 3^v \cdot \mu$

$3 \mid (p+1)$

$\Rightarrow$ cube map **3-to-1**

$\alpha,\ \alpha \cdot \zeta_3,\ \alpha \cdot \zeta_3^2$

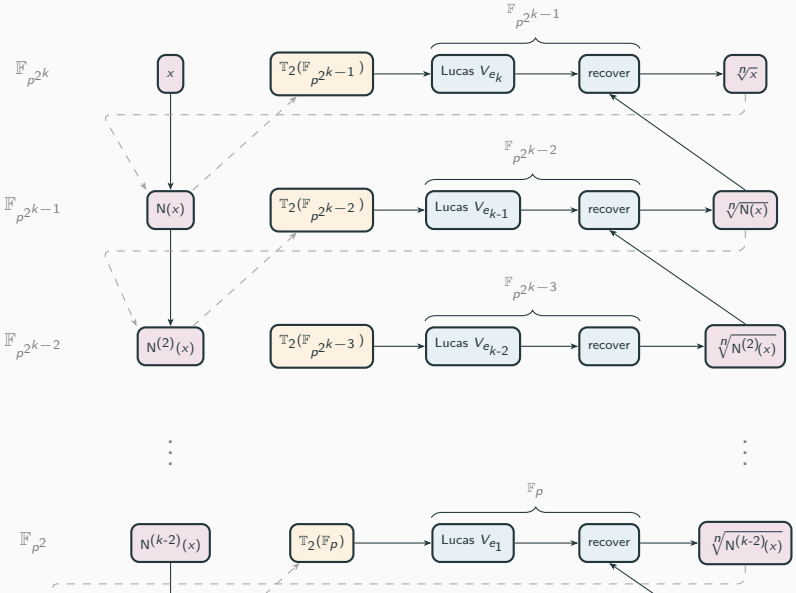
all cube to α^3 on $\mathbb{T}_2$

- Project: $\alpha = x^{p-1}$
- Rotate: $e' = 3^{-1} \bmod \mu$
  $\alpha^{e'}$; exact if $v = 1$
- Correct $\alpha^{e'} \to \alpha^e$
  3-Sylow, if $v \geqslant 2$
- Recover z_0, z_1

$\mathbb{F}_p$ cbrt unique: $a^{(2p-1)/3}$

No ζ-adjust needed

Appendix: Recursive n-th root in $\mathbb{F}_{p^{2^k}}$



odd n, $\gcd(n, p+1) = 1$

each level:

- N: drop to sub-field
- $\mathbb{T}_2$: torus + Lucas
- recover $\sqrt[n]{\cdot}$

$\downarrow$ norm recurses down

$\uparrow$ n-th root feeds back

GLV if $j \geqslant 1$